Lecture 7

Sequencing Data Formats

Genbank, FASTA and FASTQ

What is DATA?

I know DATA when I see it



What is data?

Data has two properties

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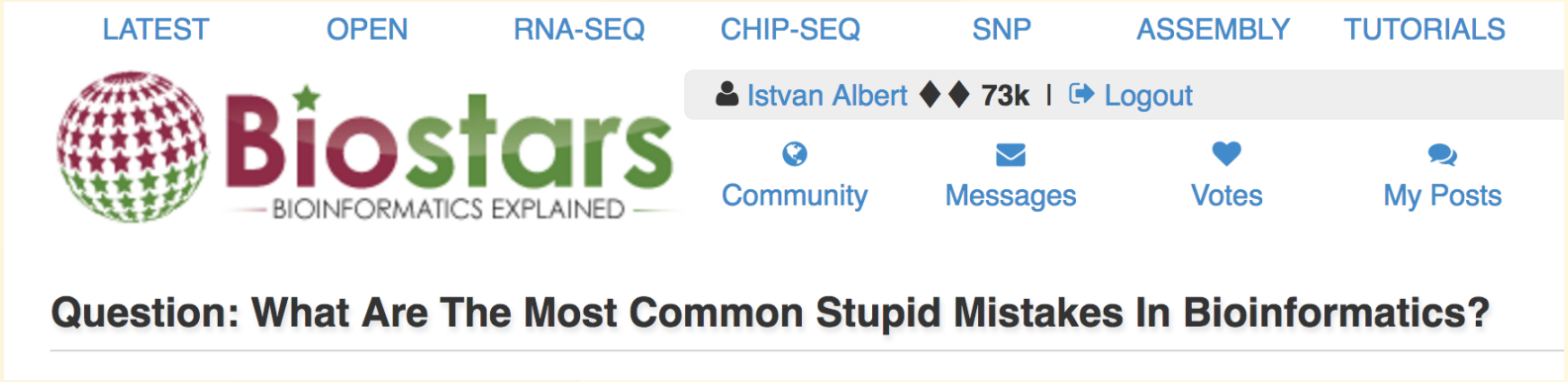
1. Data contains **some** information.
2. Data has a format (an optimization). It is designed **to allow you to do something well.**

Each **data type** is a **subset** of all existing information, optimized for a particular task.

No wonder we have so many types of data.

Enduring Love for Data Formats

Biostar Question of the Day: **What are the Most Common Stupid Mistakes in Bioinformatics?**



Highest voted answer:

> *Invent a new, weakly defined, internally redundant, ambiguous, bulky, fruit salad of a data format. Again.*

Main classes of data types

You'll run into the following data types:

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1. Sequence data: `FASTA`, `FASTQ`
2. Interval data: `BED`, `GFF`, `SAM`, `VCF`
3. Knowledge data (mini encyclopedia): `GenBank`, `EMBL`, `UniProt`
4. *Weakly defined, internally redundant, ambiguous, bulky, fruit salad* data.

There may be slight variations within a class.

Interval datasets

These may be broken down into sub types

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1. Annotations: `BED`, `GFF`
2. Alignment representation: `SAM`
3. Variation representation: `VCF`

Data formats die very slowly. A popular software tool can temporarily "resurrect" inefficient, unused and obscure formats:

Everyone thought GFF version 2 was dead until TopHat came along. Thanks, TopHat! (sarcasm...)

Why do bioinformaticians dislike formats?

You too will have this problem:

> *I have data in **Format A** but an essential process requires it in **Format B**. How do I convert it?*

Brent Pedersen on Biostar: **Very Bad Things**:

> *I've been doing bioinformatics for about 10 years now. I used to joke with a friend of mine that most of our work was converting between file formats. We don't joke about that anymore.*

How do I convert formats?

Best advice: Try to avoid converting!

Do your best to get data in the right format!

Even data types that appear to represent similar information may have content that does not fit into the other type!

Conversion software may add "extra" little features that you may be unaware of.

In simple cases use reformatters such as `readseq` or `seqret`.

Knowing your datatypes

Understanding what each data may or may not contain is a core bioinformatics skill.

- Every data type can be more complicated than what it appears.
- Scientists regularly encode more information than what the file was supposed to store.
- The rules are not always followed strictly.

And that's why we all love them data formats.

The Genbank format

The most “ancient” bio data format. Designed to be:

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1. human readable
2. reasonably complete

It used to **printed out** as a book! No really!

Typically contains pgene annotations (start, end) features and the sequence (full GenBank) as well publication references.

Getting a GenBank file

```
efetch -db nuccore -id AF086833 -format gb | more
```

shows:

```
LOCUS       AF086833                18959 bp    cRNA    lin
DEFINITION  Ebola virus - Mayinga, Zaire, 1976, complete g
ACCESSION   AF086833
VERSION     AF086833.2
KEYWORDS    .
SOURCE      Ebola virus - Mayinga, Zaire, 1976 (EBOV-May)
ORGANISM    Ebola virus - Mayinga, Zaire, 1976
            Viruses; ssRNA viruses; ssRNA negative-strand
            Mononegavirales; Filoviridae; Ebolavirus.
REFERENCE   1  (bases 1 to 18959)
```

Page through it. See what the file contains.

FASTA format

This is a single FASTA record:

```
>identifier then optional other information may go here
AAATATTAAATTAATTAATGCAATTCGAA
ATGCAATTCGAAATGCAATTCGAAATGCA
```

The sequence needs to follow an **alphabet**.

See International Union of Pure and Applied Chemistry (IUPAC) codes for nucleic acids

IUPAC Alphabets

The content of the sequence section of the record:

- Nucleotide codes: `ATGC`
- RNA codes: `AUGC`
- Amino acid codes: `ACDEFGHIKMNPQRSTVWY`
- Extended alphabets: `WSKMBDHVN.-`

Extended alphabet has amibguity codes:

- **W** weak: `A` or `T`,
- **S** strong: `C` or `G`

Getting a FASTA file

Download data for an accession number from NCBI:

```
efetch -db nuccore -id AF086833 -format fasta | more
```

will print:

```
>AF086833.2 Ebola virus - Mayinga, Zaire, 1976, complete g
CGGACACACAAAAAGAAAGAAGAATTTTTAGGATCTTTTGTGTGCGAATAACTATGAG
TTTTCCTCTCATTGAAATTTATATCGGAATTTAAATTGAAATTGTTACTGTAATCACA
CAGAGCCACATCACAAAGATAGAGAACAACCTAGGTCTCCGAAGGGAGCAAGGGCATC
TGAAAATCCCTTGTCAACACCTAGGTCTTATCACATCACAAGTTCCACCTCAGACTCT
AACAACCTTAATAGAAACATTATTGTTAAAGGACAGCATTAGTTCACAGTCAAACAAG
TTAACCTTGGTTTTGAACTTGAACACTTAGGGGATTGAAGATTCAACAACCCTAAAGC
```

FASTA: Deceiving simplicity

The FASTA format seemed to so simple that they felt unnecessary to define it.

Hence people have been trying to cram extra information into the file format ever since its invention.

Many tools (tacitly) require FASTA files to be formatted certain ways.

See the book for details.

Convert GenBank to FASTA

Note: `readseq` *is not present in bioconda yet. Install manually. See the book.* Get a GenBank file:

```
efetch -db=nuccore -format=gb -id=AF086833 > AF086833.gb
```

Convert to FASTA:

```
cat AF086833.gb | readseq -p -format=FASTA | head
```

prints:

```
>AF086833 Ebola virus - Mayinga, Zaire, 1976, complete gen
cggacacacaaaaagaaagaagaatttttaggatctttgtgtgcgaataactatgag
agattaataattttcctctcattgaaatttatatcggaatttaaattgaaattgttac
```

Note: The converted file is not identical to the one obtained from NCBI

Play the "I spy" game.

How many differences can you see?

Convert GenBank to interval (GFF)

There are different tools to do this:

```
cat AF086833.gb | readseq -p -format=GFF | head -4
```

This converts all the features in the file:

```
##gff-version 2
# seqname source feature start end score strand frame attr
AF086833 - source 1 18959 . + . organism "Ebola virus - Ma
AF086833 - 5'UTR 1 55 . + . note "putative leader region"
AF086833 - gene 56 3026 . + . gene "NP"
AF086833 - mRNA 56 3026 . + . gene "NP" ; product "nucleop
```

Does this follow the Sequence Ontology? Not really!
See: `5'UTR` vs `five_prime_utr`

The FASTQ format

Used by sequencing instruments.

If the sequences are measurements, there should be a way to associate a quality measure to each base.

The quality tells us how accurate each measurement ("base call") is.

FASTQ --> FASTA with qualities.

Extension: `.fq`, `.fastq`

The idea behind FASTQ

For each base call, the instrument assign a probabiliy that it is incorrect (1 means 100%):

```
A -> 0.00001
T -> 0.1
G -> 1
C -> 0.01
```

Write it horizontally like sequences:

```
   A        T       G       C
0.0001     0.1      1     0.01
```

Now let's try to make the bottom line the same size as the top one.

Phred Encoding

Remap numbers (via a convoluted scheme)

```
0.0001 = 1/10,000  = 1E-4  = 1E-(40/10) -> 40 -> I
0.001  = 1/100     = 1E-2  = 1E-(20/10) -> 20 -> 5
```

now:

```
   A       T      G       C
0.0001     0.1    1     0.01
   I       +      !       5
```

The sequence will be reported as:

```
ATGC
I+!5
```

FASTQ format scale

Visually the remapping scale:

```
!"#$%&'()*+,-./0123456789:;<=>?@ABCDEFGHI
|    |    |    |    |    |    |    |    |
0....5...10...15...20...25...30...35...40
|    |    |    |    |    |    |    |    |
worst.............................best
```

```
I+!5   -> 40  10  0  20
```

`40` -> one in ten thousand `1E-40/10`

`10` -> one in ten `1E-10/10`

The FASTQ format

Four lines per record

1. `@` indicates the sequence identifier
2. The sequence content of the read
3. `+` optionally repeat the sequence id (often left empty)
4. Sequence quality string

```
@data
ATGC
+
I+!5
```

Real FASTQ files are hard on eyes

Other information crammed into the sequence id.

```
@HWI-EAS209_0006_FC706VJ:5:58:5894:21141#ATCACG/1
GATTTGGGGTTCAAAGCAGTATCGATCAAATAGTAAATCCATTTGTTCAACTCACAGT
+
!''*((((****+))%%%++)(%%%%).1***-+*''))**55CCF>>>>>>CCCCCCC
```

See the handbook for details on how to parse out the header.

ID line for an Illumina instrument:

```
@HWI-EAS209_0006_FC706VJ:5:58:5894:21141#ATCACG/1
```

Eyeballing qualities

We rarely need to find out precisely what a value decodes to. The simplest is to remember that:

- `!"#$%&'()*+,-.` means low quality 1/10
- `0123456789` means medium quality 1/100
- `ABCDEFGHI` means high quality 1/1000



Tip: If the quality string looks like a swearing in comic book `$#!@@#$%*W*!!!` then it means low quality.

Things instument makers don't to talk about

The FASTQ probabilities are not accurate!

Most instruments "guesstimate" qualities!

Qualities don't correspond to actual sequence identity probabilities.

Qualities are most useful to recognize big and systematic problems during sequencing.

But instruments makers don't like to talk about things that could go wrong.